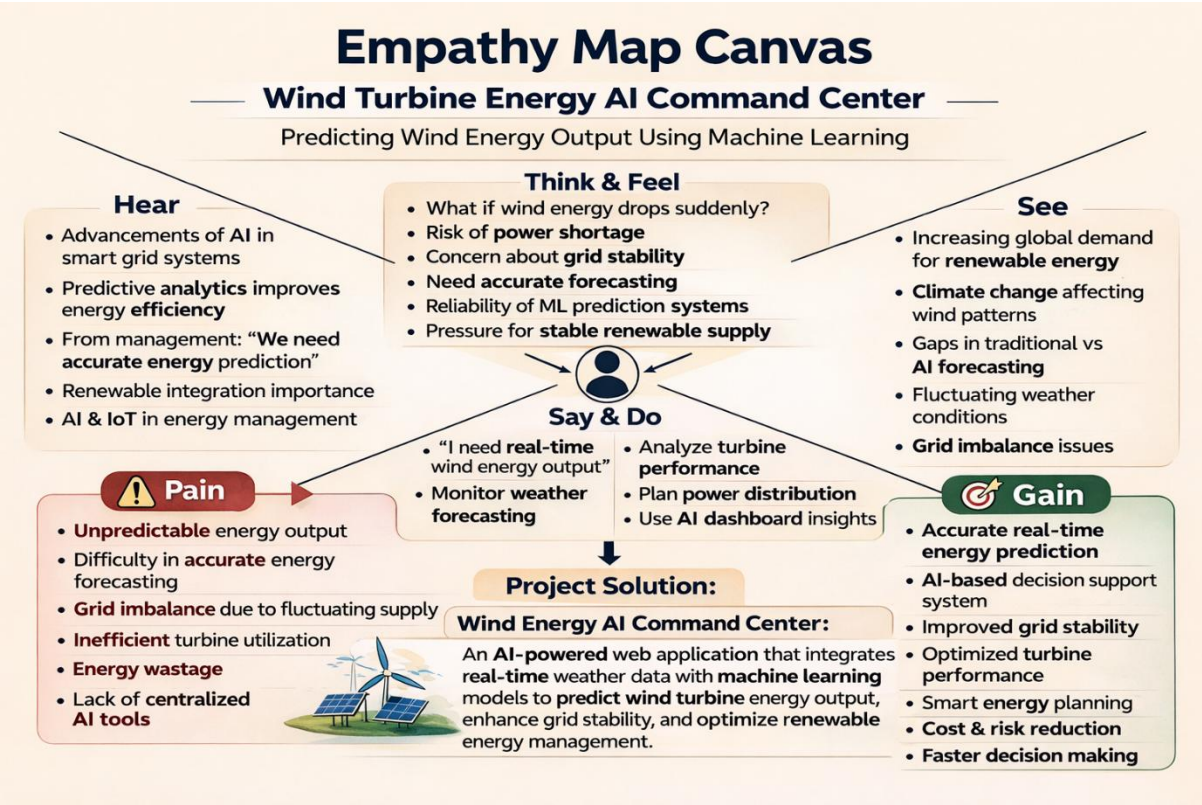
Ideation Phase
Empathize & Discover

Date	22 February 2026
Team ID	LTVIP2026TMIDS42033
Project Name	**Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management**
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple visual tool used to understand the thoughts, feelings, challenges, and needs of a wind farm operator or grid manager. It helps the team analyze real-world problems such as unpredictable wind energy output, grid instability, and forecasting difficulties. By viewing the system from the user's perspective, we can design an AI-powered solution that provides accurate real-time energy prediction, improves decision-making, and optimizes renewable energy management.

Example:



Empathy Map Canvas

Wind Turbine Energy AI Command Center

Predicting Wind Energy Output Using Machine Learning

Hear

- AI improves renewable energy forecasting
- Smart grids need **predictive** analytics
- "We need real-time energy **monitoring**
- "Reduce power **fluctuation** losses
- **Optimize** turbine performance
- Renewable integration is the future

Think & Feel

What if wind speed drops suddenly?

Need accurate forecasting

Risk of power shortage

Reduce power fluctu- losses

"I need real-time wind energy output"

Spend 30m energy monitoring

Monitor weather dashboards

Traditional forecasing limitations

Wind Farm Operator / Grid Manager

See

- Changing weather conditions
- Climate change affecting wind patterns
- Increasing demand for renewable energy
- **Grid imbalance** issues
- Traditional forecasting liimátions
- **Need for digital** energy transformation
- Energy waste due to poor planning

Cim ıuating energy output **is stressful**

Pain

- Unpredictable energy output
- Grid instability risk
- Energy wastage
- Inefficient turbine utilization
- Lack of **centralized smart** analytics
- Inaccurate forecasting dashboards
- Energy wastage
- Manual decision delays
- Lack of centralized smart analytics

Gain

- Accurate real-time energy prediction
- Increase turbine performance daily
- Lack of centralized smart analytics
- Accurate **real**-time energy prediction
- Improved grid stability
- Optimized turbine performance
- Smart renewable planning
- Faster decision making